

# Structural Diagnosability of Enterprises as Continua: Formal Definition and Decidability

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## Abstract

We address a single frozen scientific question: Is *structural diagnosability* a formally well-defined and decidable property of enterprises modeled as continua ( $K_{EA}$ ), independent of managerial intent and KPI observability?

This preprint provides a formal definition of structural diagnosability strictly within the immutable OC core and the executable-closed enterprise continuum specification (ETS.K\_EA.v1.1). We then reduce diagnosability to a finite decision problem over the enterprise continuum representation, yielding a decidability result under explicitly stated minimal assumptions already present in the specification.

**Keywords.** structural diagnosability; enterprise continuum; formal methods; decidability; OC;  $K_{EA}$

## 1 Problem and Scope

### 1.1 Frozen scientific question

This preprint answers exactly one question (Q1):

“Is structural diagnosability a formally well-defined and decidable property of enterprises modeled as continua ( $K_{EA}$ ), independent of managerial intent and KPI observability?”

### 1.2 Hard scope constraints

We enforce the following constraints:

- No new primitives beyond OC core and ETS.K\_EA.v1.1.
- No reinterpretation of OC or ETS.K\_EA.v1.1.
- No methodological or prescriptive content (no roadmaps, no consulting, no advice).
- No weakening of STOP / No-Go semantics and invariants.
- The paper stands alone; it does not require reading any whitepaper.

### 1.3 Operational meaning

The term ‘structural’ in this paper refers to structure encoded in the enterprise continuum formalism (state space, boundaries, axes, thresholds, flows, cycles, and continuity). The term ‘diagnosability’ refers to whether a decision procedure can determine, from a formally admissible description of an enterprise-as-continuum, whether the continuum is diagnosable in the sense defined in section 3.

We explicitly do *not* assume the availability of KPIs, managerial intent, or any specific human interpretation layer. Any observation model, if used, is introduced only as a derived construct definable from the immutable primitives.

## 2 Formal Setting

We work strictly within the OC core primitives and the enterprise-continuum specification  $K_{EA}$  as fixed by `ETS.K_EA.v1.1`. The enterprise continuum is treated as a formal object equipped with (at minimum) the following immutable components:

$$(\Omega(K_{EA}), \partial\Omega(K_{EA}), A(K_{EA}), P(K_{EA}), \Theta(K_{EA}), J(K_{EA}), C(K_{EA}), k(K_{EA})).$$

All symbols above refer exclusively to the corresponding definitions provided by OC core and `ETS.K_EA.v1.1`. No reinterpretation or semantic extension is permitted.

### 2.1 Representational assumption

To formulate decidability results, we require that an admissible instance of  $K_{EA}$  be provided in a *finite*, machine-readable representation consistent with `ETS.K_EA.v1.1`. This assumption does not introduce a new primitive; it merely makes explicit the input model required for any effective decision procedure.

In particular, finiteness applies to the representations of axes, thresholds, flows, cycles, and all auxiliary data structures prescribed by ETS, yielding a finite encoding domain for admissible enterprise states.

### 2.2 Permitted information for decision procedures

Any decision procedure considered in this work may inspect only the following ETS-provided structures of the given  $K_{EA}$  instance:

- the formal representation of admissible states in  $\Omega(K_{EA})$  and the associated boundary structure  $\partial\Omega(K_{EA})$ ;
- the finite sets of axes  $A(K_{EA})$  and thresholds  $\Theta(K_{EA})$ ;
- any explicitly represented flow and cycle data  $(J(K_{EA}), C(K_{EA}))$  contained in the ETS-conformant input;
- the STOP/No-Go constraints and invariants encoded in `ETS.K_EA.v1.1`.

No appeal to managerial intent, KPI observability, or external interpretive semantics is permitted.

### 2.3 Structure of the formal argument

Within this setting, we proceed as follows: first, structural diagnosability is defined as a Boolean property determined solely by OC and ETS structures; second, this property is shown to be well-defined (invariant under ETS-equivalent representations); and third, decidability is established by reduction to a finite decision problem over the ETS-conformant representation.

## 3 Definitions

**Definition 1** (Admissible enterprise-continuum instance). An *instance* of  $K_{EA}$  is an input object that conforms to **ETS.K\_EA.v1.1**. Such an instance determines, without reinterpretation or semantic enrichment, the corresponding OC core structures  $\Omega(K_{EA})$ ,  $\partial\Omega(K_{EA})$ ,  $A(K_{EA})$ ,  $\Theta(K_{EA})$ ,  $J(K_{EA})$ ,  $C(K_{EA})$ , and  $k(K_{EA})$ , exactly as specified by OC core and **ETS.K\_EA.v1.1**.

**Definition 2** (ETS-induced diagnostic predicates). Fix an admissible instance of  $K_{EA}$ . Let  $\text{Enc}(K_{EA})$  denote the finite set of ETS-admissible encodings of enterprise states for this instance, as determined by the finite representations of axes, potentials, thresholds, flows, and cycles prescribed by **ETS.K\_EA.v1.1**.

The *diagnostic predicate family*  $\text{Pred}_D(K_{EA})$  is the finite set of Boolean predicates over  $\text{Enc}(K_{EA})$  defined as follows:

1.  $\text{In}_\Omega(x)$ : evaluates to true iff the state decoded from  $x$  is an element of  $\Omega(K_{EA})$ .
2.  $\text{On}_{\partial\Omega}(x)$ : evaluates to true iff the state decoded from  $x$  lies on the boundary  $\partial\Omega(K_{EA})$ .
3.  $\text{Sat}_\Theta(x)$ : evaluates to true iff the decoded state satisfies all threshold constraints specified by  $\Theta(K_{EA})$  under OC+ETS semantics.
4.  $\text{Val}_k(x)$ : evaluates to true iff the continuity measure  $k(K_{EA})$  is well-defined at the decoded state and admits evaluation in the numeric domain prescribed by **ETS.K\_EA.v1.1**.

All predicates in  $\text{Pred}_D(K_{EA})$  are defined solely in terms of OC core structures and **ETS.K\_EA.v1.1**. No predicate may reference intent, KPI signals, interpretation layers, or external annotations.

**Definition 3** (Structural diagnosability). Let  $K_{EA}$  be an admissible enterprise-continuum instance with finite encoding set  $\text{Enc}(K_{EA})$  and predicate family  $\text{Pred}_D(K_{EA})$ .

The enterprise continuum  $K_{EA}$  is said to be *structurally diagnosable* iff for every predicate  $P \in \text{Pred}_D(K_{EA})$  and every encoding  $x \in \text{Enc}(K_{EA})$ , the truth value  $P(x)$  is well-defined and computable under the OC+ETS.K\_EA.v1.1 semantics of the instance.

Structural diagnosability is encoded as the Boolean property

$$D(K_{EA}) = \begin{cases} 1, & \text{if all } P(x) \text{ are well-defined and computable,} \\ 0, & \text{otherwise.} \end{cases}$$

*Remark 1* (Operational form). Since both  $\text{Enc}(K_{EA})$  and  $\text{Pred}_D(K_{EA})$  are finite,  $D(K_{EA}) = 1$  is equivalent to the existence of a total decision procedure that, given any  $P \in \text{Pred}_D(K_{EA})$  and  $x \in \text{Enc}(K_{EA})$ , halts and returns the correct truth value of  $P(x)$ .

*Remark 2* (Independence from intent and KPI observability). Independence from managerial intent and KPI observability is enforced by construction. The definition of  $D(K_{EA})$  depends exclusively on ETS-induced predicates over OC core structures. No intent variables, KPI signals, or interpretive layers are admissible as inputs.

## 4 Main Results

**Theorem 1** (Well-definedness of structural diagnosability). *Let  $K_{\text{EA}}$  be an admissible enterprise-continuum instance. The Boolean property  $D(K_{\text{EA}})$  is well-defined in the following sense: if two concrete representations  $R_1$  and  $R_2$  are equivalent under  $\text{ETS.K\_EA.v1.1}$ , then they induce the same value of  $D(K_{\text{EA}})$ .*

*Proof sketch.* ETS-equivalent representations induce the same OC-core objects:  $\Omega(K_{\text{EA}})$ ,  $\partial\Omega(K_{\text{EA}})$ ,  $A(K_{\text{EA}})$ ,  $\Theta(K_{\text{EA}})$ ,  $J(K_{\text{EA}})$ ,  $C(K_{\text{EA}})$ , and  $k(K_{\text{EA}})$ . Consequently, they induce the same finite encoding domain  $\text{Enc}(K_{\text{EA}})$  and the same diagnostic predicate family  $\text{Pred}_D(K_{\text{EA}})$ .

The value of  $D(K_{\text{EA}})$  depends exclusively on whether all predicate evaluations  $P(x)$ , for  $P \in \text{Pred}_D(K_{\text{EA}})$  and  $x \in \text{Enc}(K_{\text{EA}})$ , are well-defined and computable under the fixed OC/ETS semantics. Since both the domain and semantics are invariant under ETS-equivalent representations, the truth of this condition is invariant as well. Therefore  $D(K_{\text{EA}})$  is representation-independent.  $\square$

**Theorem 2** (Decidability of structural diagnosability). *Let an admissible  $K_{\text{EA}}$  instance be given by a finite  $\text{ETS.K\_EA.v1.1}$ -conformant representation. Then the structural diagnosability property  $D(K_{\text{EA}})$  is decidable.*

*Proof sketch.* By the representational assumption, both the encoding set  $\text{Enc}(K_{\text{EA}})$  and the predicate family  $\text{Pred}_D(K_{\text{EA}})$  are finite.

Each predicate  $P \in \text{Pred}_D(K_{\text{EA}})$  is defined by explicit, algorithmically specified OC/ETS rules (membership in  $\Omega$ , boundary checks for  $\partial\Omega$ , threshold satisfaction, and evaluation of  $k$  in the prescribed numeric domain). Therefore, for any fixed  $x \in \text{Enc}(K_{\text{EA}})$ , the computation of  $P(x)$  either terminates with a Boolean value or fails to be well-defined.

To decide  $D(K_{\text{EA}})$ , one enumerates all pairs  $(P, x) \in \text{Pred}_D(K_{\text{EA}}) \times \text{Enc}(K_{\text{EA}})$  and checks whether each evaluation  $P(x)$  is well-defined and computable. Since this is a finite set of checks over finite data, the procedure terminates. Consequently,  $D(K_{\text{EA}})$  is decidable, as a standard finite decision problem over a finite encoding domain Sipser 2012; Clarke et al. 2018.  $\square$

## 5 Scope Limits and Non-Claims

### 5.1 What is not claimed

We do not claim:

- that diagnosability implies controllability, success, or improvement;
- any managerial prescription or actionability;
- any empirical KPI-based observability guarantee;
- any universality beyond the scope of OC and  $\text{ETS.K\_EA.v1.1}$ .

### 5.2 What would falsify the main claims

The main claims would be falsified if:

- the formal definition of  $D(K_{\text{EA}})$  cannot be expressed purely in the immutable structures provided by OC and  $\text{ETS.K\_EA.v1.1}$ ; or
- the ETS representation necessarily requires infinite objects that are not reducible to finite decision procedures in the intended encoding.

## References

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- Sipser, Michael (2012). *Introduction to the Theory of Computation*. 3rd ed. Cengage Learning.